

1 Problem Description

There is a line segment with end points $(1, 2)$ and $(12, a)$ where $a > 2$.

The line segment is rotated around the x-axis, making a frustrum f_x .

The line segment is rotated around the y-axis, making a frustrum f_y .

The difference between the volumes of these 2 frustrums is 50 cubic units.

Find the sum of all the possible values of a

2 Problem Solution

2.1 Modelling the line segment

To do this I will treat this problem as a volumes of revolution problem.

I will do this by modelling the line segment as a line and when taking the integral I will set the limits to the line segments limits.

Finding gradient

$$\begin{aligned} m &= \frac{a-2}{12-1} \\ &= \frac{a-2}{11} \end{aligned}$$

finding equation of line, y^2 and x^2

$$\begin{aligned} \Rightarrow y - 2 &= \frac{a-2}{11}(x-1) \\ \Rightarrow y &= \frac{a-2}{11}(x-1) + 2 \\ \Rightarrow y^2 &= \left(\frac{a-2}{11}(x-1) + 2 \right)^2 \\ &= \frac{(a-2)^2}{121}(x-1)^2 + \frac{4(a-2)}{11}(x-1) + 4 \\ &= \frac{(a-2)^2}{121}(x^2 - 2x + 1) + \frac{4(a-2)}{11}(x-1) + 4 \\ \Rightarrow x &= \frac{11(y-2)}{a-2} + 1 \\ \Rightarrow x^2 &= \frac{121(y-2)^2}{(a-2)^2} + \frac{22(y-2)}{a-2} + 1 \\ &= \frac{121}{(a-2)^2}(y^2 - 4y + 4) + \frac{22}{a-2}(y-2) + 1 \end{aligned}$$

finding fulcrum from rotation around x-axis (f_x)

$$\begin{aligned}
& \pi \int_1^{12} \left(\frac{(a-2)^2}{121} (x^2 - 2x + 1) + \frac{4(a-2)}{11} (x-1) + 4 \right) dx \\
&= \frac{(a-2)^2}{121} \pi \int_1^{12} (x^2 - 2x + 1) dx + \frac{4(a-2)}{11} \pi \int_1^{12} (x-1) dx + \pi \int_1^{12} 4 dx \\
&= \frac{(a-2)^2}{121} \pi \left[\frac{x^3}{3} - x^2 + x \right]_1^{12} + \frac{4(a-2)}{11} \pi \left[\frac{x^2}{2} - x \right]_1^{12} + \pi [4x]_1^{12} \\
&= \frac{(a-2)^2}{121} \pi \left(\frac{12^3}{3} - 12^2 + 12 - \left(\frac{1^3}{3} - 1^2 + 1 \right) \right) \\
&\quad + \frac{4(a-2)}{11} \pi \left(\frac{12^2}{2} - 12 - \left(\frac{1^2}{2} - 1 \right) \right) \\
&\quad + \pi(4 \times 12 - 4 \times 1) \\
&= \frac{11}{3} \pi (a-2)^2 + 22(a-2)\pi + 44\pi
\end{aligned}$$

finding fulcrum from rotation around y-axis (f_y)

$$\begin{aligned}
f_y &= \pi \int_2^a \left(\frac{121}{(a-2)^2} (y^2 - 4y + 4) + \frac{22}{a-2} (y-2) + 1 \right) dy \\
&= \frac{121}{(a-2)^2} \pi \int_2^a (y^2 - 4y + 4) dy + \frac{22}{a-2} \pi \int_2^a (y-2) dy + \pi \int_2^a 1 dy \\
&= \frac{121}{(a-2)^2} \pi \left[\frac{y^3}{3} - 2y^2 + 4y \right]_2^a + \frac{22}{a-2} \pi \left[\frac{y^2}{2} - 2y \right]_2^a + \pi [y]_2^a \\
&= \frac{121}{(a-2)^2} \pi \left(\frac{a^3}{3} - 2a^2 + 4a - \left(\frac{2^3}{3} - 2 \times 2^2 + 4 \times 2 \right) \right) \\
&\quad + \frac{22}{a-2} \pi \left(\frac{a^2}{2} - 2a - \left(\frac{2^2}{2} - 2 \times 2 \right) \right) \\
&\quad + \pi(a-2) \\
&= \frac{121}{(a-2)^2} \pi \left(\frac{a^3}{3} - 2a^2 + 4a - \frac{8}{3} \right) + \frac{22}{a-2} \pi \left(\frac{a^2}{2} - 2a + 2 \right) + \pi(a-2) \\
&= \frac{121}{3(a-2)^2} \pi (a^3 - 6a^2 + 12a - 8) + \frac{22}{2(a-2)} \pi (a^2 - 4a + 4) + \pi(a-2) \\
&= \frac{121}{3(a-2)^2} \pi (a-2)^3 + \frac{11}{a-2} \pi (a-2)^2 + \pi(a-2) \\
&= \frac{121}{3} \pi (a-2) + 11\pi(a-2) + \pi(a-2) \\
&= (a-2)\pi \left(\frac{121}{3} + 11 + 1 \right) \\
&= \frac{157\pi(a-2)}{3}
\end{aligned}$$

Now say $f_x \pm 50 = f_y$ are equal with a differerance of 50

$$\frac{11}{3}\pi(a-2)^2 + 22(a-2)\pi + 44\pi = \frac{157\pi(a-2)}{3} \pm 50$$

Let $z = a - 2$

$$\frac{11}{3}\pi z^2 + 22z\pi + 44\pi = \frac{157\pi z}{3} \pm 50$$

Now divide by pi and multiply by 3

$$11z^2 + 66z + 132 = 157z \pm \frac{150}{\pi}$$

$$11z^2 - 91z + 132 \mp \frac{150}{\pi} = 0$$

$z = 7.210464$ and 1.062263 and 5.013241 and 3.25948

$$\begin{aligned} \sum a &= 16.545 + 8 \\ &= 24.545 \end{aligned}$$